Historical Earnings Moves, ATM Straddle Pricing, and Accuracy of Post-Earnings Moves

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August 2026

1 Abstract

This paper begins with utilizing roughly 20,000 SEC EDGAR earnings events across all S&P 500 constituents from 2016-2026, to observe patterns among historical realized earnings moves and examine its forecasting ability for future earnings events. The typical post-earnings move across the entire dataset is a modest 3.1%. From an index perspective, on average a stock's two year post-earnings realized history tells you, within a 4% margin, how much it will move on the next earnings date. While any single stock may miss the mark on any given day, across 502 securities and 10 years, the individual misses balance out. A stock's realized history is found to stay consistent across time – stocks which tend to be volatile post-earnings remain volatile – but is of no use in predicting a future move on a specific event basis. The second part of the paper seeks to examine how correlated a stock's realized earnings history is to the options market's implied expected move as priced into an ATM straddle. Furthermore, the study seeks to observe and analyze patterns between the two predictive measures, that is historical earnings moves and the option market's implied expected moves, and compare the predictive parameters to the realized post-earnings move during a live earnings season. With 133 live pre-earnings ATM straddle prices, collected over the 2026 Q2 season, the analysis presented 53% ($R^2 = 0.53$) of options pricing was driven entirely by the stock's 8-quarter historical earnings mean. The Spearman correlation (ranked correlation) was slightly stronger, $\rho = 0.62$. Interestingly, having no insight into the realized moves, the market tended to under-price high-volatility stocks and over-price low-volatility stocks, relative to their historical means. Once realized moves data was extracted, it was found that when the options market priced a move well above the stock's history, the realized outcome generally validated market expectations. The live study indicated the options market charged a 7% earnings volatility risk premium. This provides evidence, albeit from a single earnings season, consistent with the well-documented finding that implied volatility trades at a premium ahead of earnings announcements, potentially providing an edge to option sellers during this period. Overall, the results of this paper suggest historical earnings behavior provides a dependable baseline for earnings risk, while option prices incorporate additional event-specific risk and, on average, charge a premium for bearing the uncertainty surrounding future earnings outcomes.

2 Introduction

As major company earnings reporting comes to a close, an interesting question regarding how well past performance dictates future results is worth considering. Financial markets are inherently unpredictable, but historical data provides the foundation for identifying recurring patterns. The critical question is how reliable these patterns are when assessing future outcomes. This paper seeks to address the following research question: *What is the correlation between a stock's historical post-earnings move and the options market's implied move? More importantly, how do live earnings outcomes perform when implied move diverges significantly from historical move?*

3 Background & Definitions

3.1 The Straddle Option

Within the Black-Scholes framework, an option's price is a direct function of expected future volatility, so option prices reveal the market's volatility expectations. In other words, the market's expectation of the range within which a security may fluctuate over the life of an option is reflected in its price. The at-the-money (ATM) straddle provides one of the cleanest and most direct estimates of the expected magnitude of that movement. An ATM straddle involves the simultaneous purchase of a call and put with the same strike price and expiration date, where the strike is nearest to the underlying security's current market price. Around earnings announcements, the price of the ATM straddle prior to release of earnings of the underlying security, serves as the option market's most direct implied estimate of the security's move post earnings, assuming the expiration and earnings release dates are nearby.

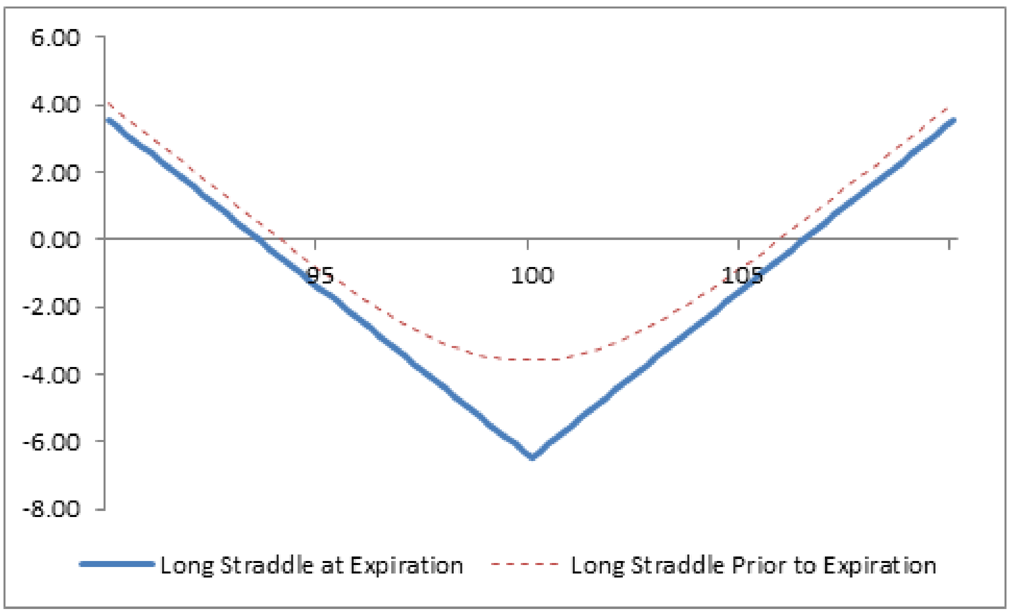

Figure 1: Long ATM Straddle Payoff Diagram

As illustrated in Figure 1, the payoff structure of a long ATM straddle features two break-even points at expiration:

$$K - (C + P)$$

and

$$K + (C + P),$$

where K denotes the strike price, C the premium paid for the call option, and P the premium paid for the put option. At expiration the underlying security's market price must surpass either point for the trader to see any profit. The total premium paid for the long ATM straddle is therefore:

$$\Pi_{\text{straddle}} = C + P. \tag{1}$$

For the purposes of this study the straddle position is not of interest from a trading perspective. Rather the focus lies on estimating the options market's implied expected move calculated by dividing the ATM straddle price by the spot price. It is this interpretation which serves as the foundation for the remainder of this paper.

3.2 Implied Expected Move

As briefly mentioned, implied expected move refers to the estimated price range of an asset derived entirely from the options market. It takes into account the expected volatility over the remaining life of the **option**; consequently, the nearer the option's expiration date is to the earnings announcement, the more directly the straddle price reflects the volatility associated with the event. If an earnings announcement occurs during a given week, but the selected option does not expire for another month, the straddle price is a poor indicator for the market's implied expected move post earnings. This is simply because the option's premium reflects the expected volatility over its entire remaining life, including a substantial amount of non-earnings-related volatility following the announcement.

Implied expected move may be presented either scaled or unscaled. Scaled versions incorporate historical volatility ratios, event-specific adjustments or other statistical multipliers. For the purposes of this study, the unscaled implied expected move is used throughout, defined as the ratio of the at-the-money (ATM) straddle price to the underlying stock price:

$$EM_{\text{implied}} = \frac{\Pi_{\text{straddle}}}{S}. \tag{2}$$

The use of an unscaled EM_{implied} is not arbitrary. Comparing current, options-driven expected moves with historical expected moves, which are inherently unscaled, ensures consistency between the theoretical quantities under comparison.

3.3 Historical Expected Move

The historical expected move is the median across the last 8-quarters of realized absolute earnings moves for any given security, computed strictly prior to each event.

$$\text{Historical EM}_t = \text{Median}\left(|R_{t-8}|, \ldots, |R_{t-1}|\right) \tag{3}$$

The computation includes the median of realized earnings moves, as opposed to the mean because then an outlier won't disrupt the overall estimate. In other words, rather than computing the average we compute the *typical* earnings reaction. This distinction is extremely pertinent when discussing specific findings in later sections of this report. As for the number of trailing events, an 8-quarter median, rather than 4-quarter or 12-quarter, was found to have a measured balance of recency and number of observations.

3.4 Realized Move

Realized move is defined as the absolute percentage change in a stock's closing price from the last unaffected trading day to the close price at the end of the first full reaction day. The formula is as follows:

$$\text{Realized Move}_t = \left| \frac{P_t}{P_{t-1}} - 1 \right| \tag{4}$$

where P_t denotes the closing price on the earnings reaction day, P_{t-1} denotes the closing price on the previous trading day, and $|\cdot|$ denotes the absolute value operator, thereby measuring the magnitude of the earnings reaction irrespective of direction. Depending on whether earnings are released before-market-open (BMO), intraday, or after-market-closes (AMC), the reaction date, and therefore the realized move computation, adjusts accordingly. For example, JPMorgan Chase & Co (JPM) released earnings on July 14th BMO, and therefore the reaction day was on the **same** day as the release date.

4 Data & Methodology

4.1 Universe & Data Sources

The equity universe used for this study comprises 502 S&P 500 constituents. The study applies the *current* S&P 500 membership to a decade of historical events, and therefore reflects survivorship bias: companies that were dropped from the period - through acquisition or financial distress - are excluded, even though they were index members when their earnings occurred. Because de-listed and acquired firms are more likely to exhibit unusually large earnings reactions, restricting the sample to surviving S&P 500 constituents likely biases the observed distribution of earnings moves modestly downward. This is primarily a bias in magnitude rather than in the relationships of interest. Since Sections 5 and 6 examine the relationship between historical and realized earnings moves, there is little reason to expect survivorship alone to systematically distort the study's principal findings. All conclusions drawn in Section 5 utilize all stocks in the S&P, while Section 6 draws on live straddle snapshots collected daily through the late-July-to-early-August earnings season, yielding roughly N clean events after liquidity and expiry filters.

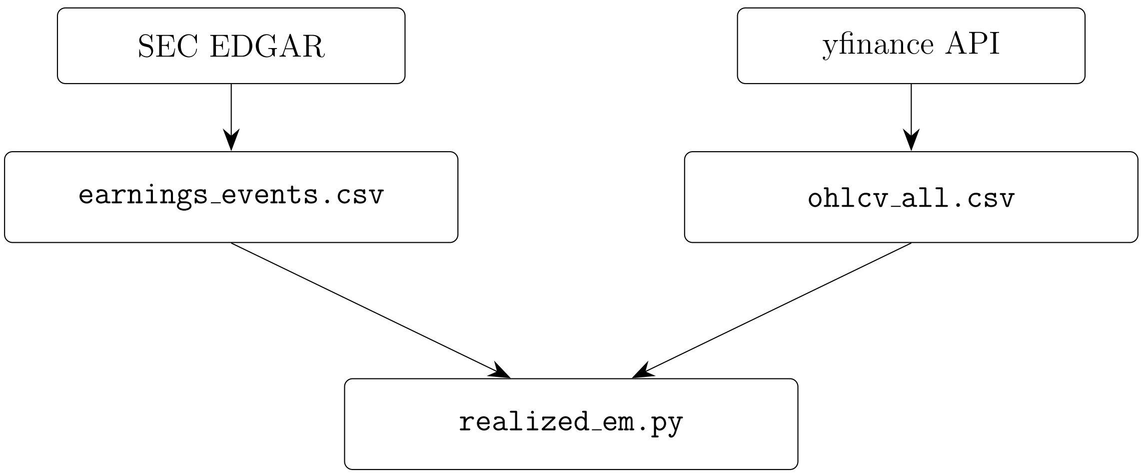

Figure 2: Historical expected move data pipeline.

The earnings event data originate from the SEC EDGAR database. The extraction process, which retrieves all Item 2.02 Form 8-K filings from January 2016 through mid-July 2026 and assigns a preliminary timing classification (BMO, Intraday, or AMC) from the filing timestamp — a classification Section 4.2 revisits. The direct output of that script is earnings_events.csv which acts as the input dataset for this study. Due to the fact that the data extraction pipeline was created as a part of a separate project, only the resulting earnings_events.csv file was incorporated into this research instead of duplicating the entire preprocessing logic. Across roughly 10 years, earnings_events.csv holds just over 21,000 events. Similarly, the logic associated with pulling open-high-low-close-volume (OHLCV) data across the last 10 years (2016-2026) is held in a separate file but outsourced originally from yfinance.

Figure 2 depicts the historical data pipeline underlying Section 5; the parallel live-collection pipeline that supplies the implied expected move is described in Section 6. The remaining sections of section 4 details the specific process going from 21,186 unadjusted and raw filings to curating a table where every event has both its realized move and the historical forecast that preceded it.

4.2 Event Identification & Reaction Day

The data sourcing from SEC EDGAR alone does not provide sufficient information to measure an earnings reaction. The exact reaction day post-earnings for all securities depend on a variety of factors and cannot be accurately concluded without greater analysis. For starters, earnings are mostly released either BMO or AMC, with intraday releases being extremely rare. The raw data, once sourced, had originally claimed 54% of earnings releases were intraday, as it was using the EDGAR filing stamps rather than the announcement times (i.e. those events were overwhelmingly pre-open (BMO) reporters whose 8-K happened to be filed mid-morning). After reclassifying the data based on correct metrics, it turns out there were no real intraday releases. A similar complication occurred where earnings were released BMO but filed AMC and were therefore listed as AMC. The 19 ambiguous cases were resolved via price-based validation - essentially viewing the absolute returns on both days in question and the day which had a substantially greater return was the actual reaction day.

4.3 Data Cleaning

With the data sourcing complications taken care of there were two remaining issues in the dataset which required cleaning. To begin, given every calendar day is not a trading day (i.e. weekends and holidays are omitted) a process called *snapping* was implemented to ensure accuracy among companies releasing earnings AMC. Rather than mechanically defining the reaction day as the calendar day following the earnings announcement, the reaction date is assigned according to the timing of the release:

```
if timing == "BMO":
    effective_date = filing_date

elif timing == "AMC":
    effective_date = filing_date + 1 calendar day

elif timing == "INTRADAY":
    effective_date = filing_date

reaction_date = next_trading_day_on_or_after(effective_date)
```

Listing 1: Reaction-date assignment based on earnings-release timing.

The latter stage of preprocessing requires an additional data-cleaning step: the de-duplication of Item 2.02 Form 8-K filings, as not every such filing corresponds to a quarterly earnings announcement. For example, Tesla, Inc. (TSLA) routinely files Item 2.02 Form 8-K reports for vehicle delivery figures, production updates, and other operational announcements. Although these filings satisfy the SEC reporting requirements, they do not represent quarterly earnings releases and, if left unfiltered, would result in duplicate earnings events within the dataset. A 30-day cleaning rule was implemented filtering out any events which occurred in multiplicity over a 30-day window. This resulted in 5% of 8-K filings being removed as non-earnings operational filings, leaving 20,099 genuine earnings events.

With the timing corrected, non-earnings filings removed, and the dataset fully cleaned, one is ready to proceed to the analysis stage. By applying the definitions laid out in Section 3, the final dataset of 20,099 scored events carries both a realized move and a trailing historical expected move. Section 5 examines what a sole reliance on history for forecasting truly reveals.

5 Part 1 - Historical Patterns

The purpose of this section seeks to identify informative patterns and observations within historical earnings data and assess how effectively past earnings behavior can forecast future earnings events.

5.1 The Distribution of Realized Earnings Moves

This subsection examines the distribution of the absolute post-earnings price movements, as a percentage return, across all securities in the cleaned dataset. By graphing the distribution of this raw data one is able to assess where the majority of observations lie, the spread and shape of the data, and identify

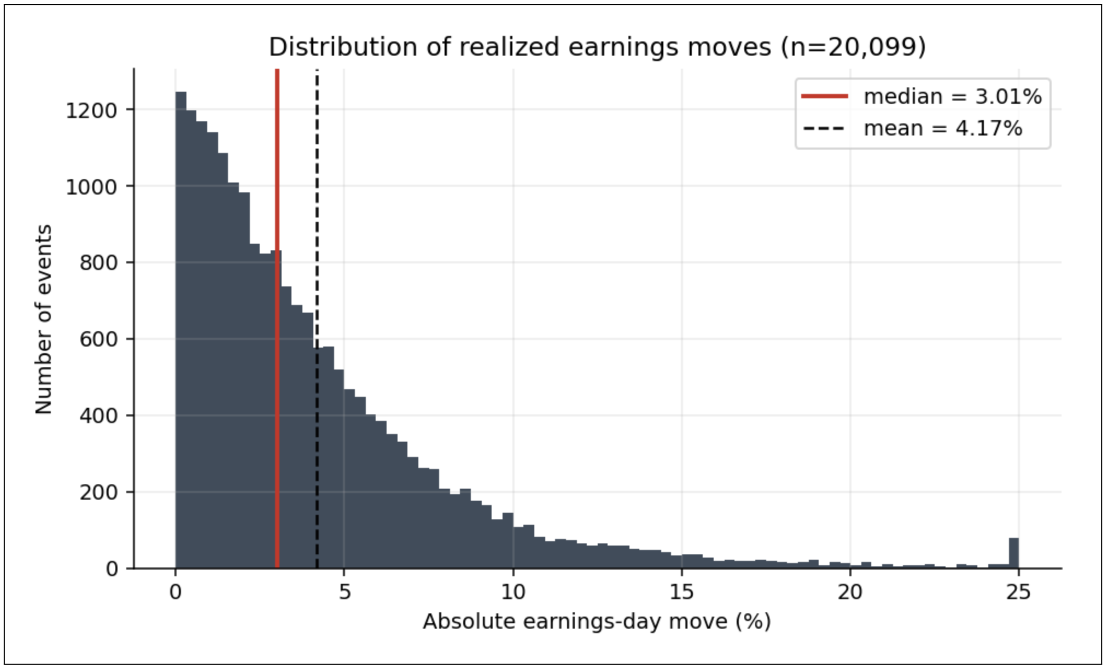

Figure 3: Histogram distribution of absolute returns across all dataset events.

any outliers. Figure 3, a histogram distributing raw absolute reaction day moves across 20,099 distinct events over 10 years, is displayed below.

As displayed one can observe the median and mean of the realized post-earnings move, is 3.01% and 4.17%, respectively. Therefore over the last 10 years across 20,099 events the *typical* post-earnings reaction move is 3.01%. Out of all post-earnings moves, the average is shown to be 4.17%, slightly higher than the median figure. The positive difference between the mean and median figures makes sense as the mean weighs extreme reaction days where stocks move significantly, while the median stays unaffected by this. Furthermore, given the ATM straddle prices the mean absolute move rather than the median, the appropriate historical benchmark for each stock in Section 6 is its own trailing mean of realized moves — not its median.

A few other observations to note regarding the graph: the 95th percentile is approximately 12.06% and the rightmost bar at the 25% level takes into account all absolute percentage moves greater than or equal to 25%, hence the magnitude of the bar.

5.2 Is History Biased?

This part of Section 5 compares the historical expected move (HEM) and what history predicted, with the realized move, and what actually happened. HEM is defined as the median of the stock's trailing 8 quarterly absolute earnings-day moves. HEM is not constant; it is a rolling measure with the eight-quarter window updating for each subsequent earnings event. For example, Apple (AAPL)'s HEM at its 2020 earnings uses post-earnings data between 2018-2020; at its 2024 earnings it uses data between 2022-2024 - sliding forward with each event. The figure below displays the graphical distribution of the ratio of realized move and trailing 8-quarter median EM across 16,101 events. As one can see the median of those ratios is approximately 1.043, very close to being unbiased.

The entire exercise of this subsection is to analyze whether a stock's past earnings moves are a fair predictor of the next one systematically. What this means is that across a 10 year period, 502 securities and 16,101 distinct earnings events the typical realized move was about 4.3% larger than its historical forecast. Unfortunately this statistic says nothing about any single event, and is therefore useless on

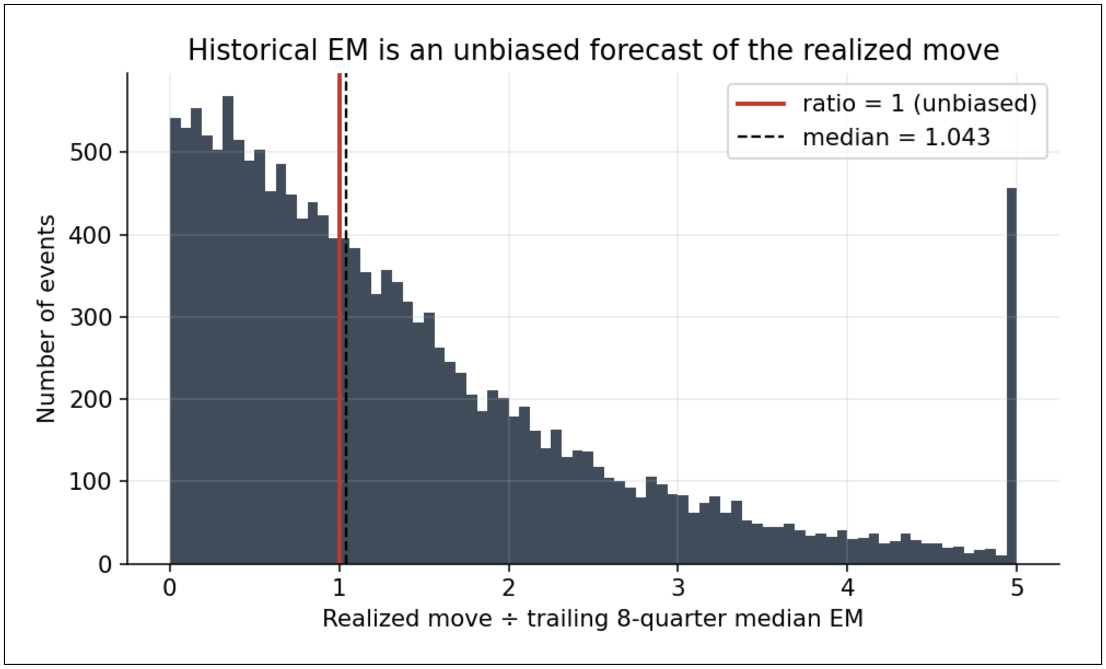

Figure 4: Histogram distribution of realized-to-forecast ratio

an event-to-event basis. Now a question worth pondering upon is whether the magnitude of a given stock's typical earnings move persists over long periods of time. In other words, is a stock's typical earnings-move size a stable trait - does a stock that moved a lot historically keep moving a lot, and a calm stock stay calm?

5.3 Do Typical Post-Earnings Moves Remain Consistent?

Persistence in typical post-earnings moves is assessed by splitting each stock's earnings history at its midpoint and comparing the median absolute post-earnings move in the first half with that of the second half. The stronger the correlation between the two, the greater the conviction that a stock's move level is a persistent characteristic rather than a transient one. Computed across all 502 stocks, the rank correlation is strong (Spearman $= 0.629$). Figure 5 shows the points clustering along the 45° line, indicating that stocks which historically moved little continue to move little, and volatile reporters remain volatile.

A spearman correlation measures whether two things move together in the same **rank order**. Therefore it asks: *if you rank the stocks by how much they moved during the first half of their history, do they stay in roughly the same rank order when ranked by the second half?* The answer to that question is for the most part, yes! Therefore one can conclude that stocks, for the most part, that historically exhibited larger earnings moves continue to do so over time, while those with smaller historical earnings moves remain relatively small. Figure 6 is presented below, representing the typical earnings move across all 11 Global Industry Classification Standard (GICS) sectors.

The final subsection of Section 5 turns to the question of greatest practical relevance to traders: *how much predictive power does a stock's historical earnings data have for its next earnings event?*

5.4 Does History Predict Timing?

This subsection deals with arguably the most important question of this section: *Can history predict, on an event-by-event basis, whether a specific upcoming earnings will move more or less than that stock's own usual level?* Section 5.3 has already established that a stock's typical post-earnings move persists

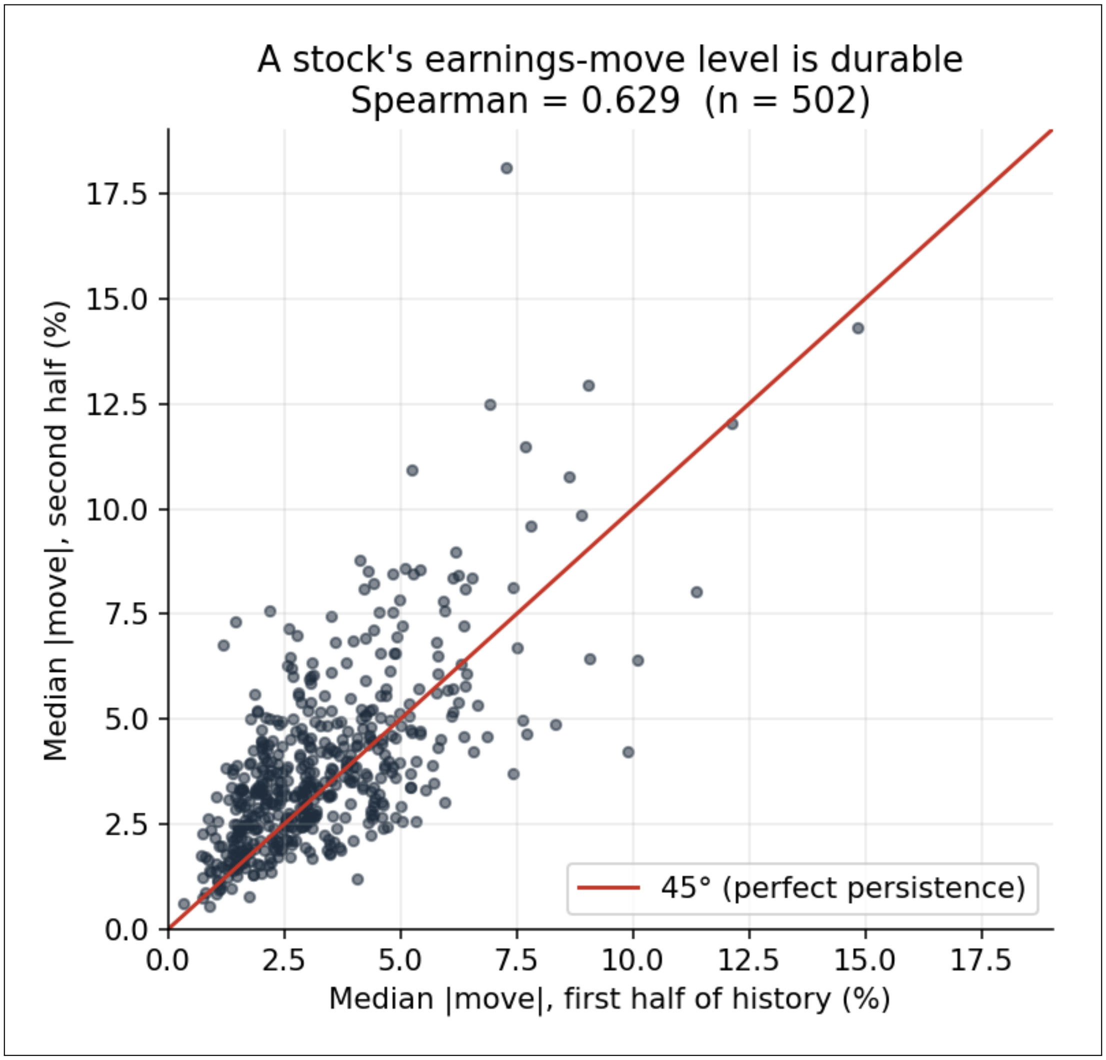

Figure 5: Persistence of earnings-move level. Each point is one stock; axes show its median absolute earnings move in the first versus second half of its history (Spearman = 0.629, n = 502). Clustering along the 45° line indicates that a stock's typical move size is durable over time.

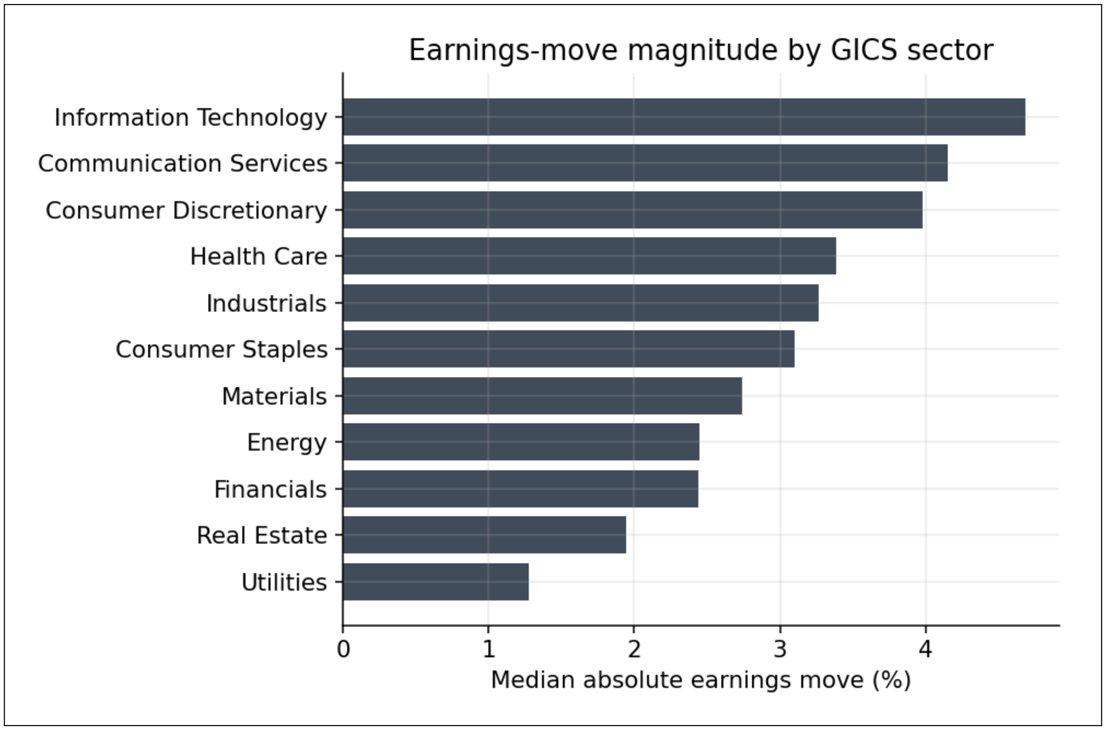

Figure 6: The median absolute earnings move per sector computed across the entire dataset.

over a period of time. But given you already know the baseline typical post-earnings move, can history narrow down, on an individual basis, which earnings event will see a deviation from that norm? A few definitions to get right prior to explaining the logic (definitions 1 and 2 have already been established):

1. **Realized earnings move.** For stock i at earnings event t, define the realized earnings move as

$$R_{i,t} = \left| \frac{P_{i,t}}{P_{i,t-1}} - 1 \right|,$$

   where i indexes the stock and t indexes the earnings event.

2. **Historical expected move (HEM).** Prior to earnings event t, define the historical expected move as the trailing eight-quarter median absolute earnings move,

$$\mathrm{HEM}_{i,t} = \mathrm{Median}\left(|R_{i,t-8}|, \ldots, |R_{i,t-1}|\right).$$

   This quantity is recomputed before every earnings announcement and therefore varies across events.

3. **Company baseline.** Define each company's long-run baseline historical expected move and realized earnings move as

$$\overline{\mathrm{HEM}}_i = \mathrm{Median}_t\left(\mathrm{HEM}_{i,t}\right),$$

$$\overline{R}_i = \mathrm{Median}_t\left(R_{i,t}\right),$$

   where the median is taken over all earnings events t for stock i within the cleaned SEC EDGAR dataset (Note: this dataset spans 10 years including roughly 40 earnings events per stock, with exceptions being tied to if a company IPO'd after 2016). These quantities produce a single baseline value for each stock.

4. **Within-stock deviations.** The median-centred variables are defined as

$$\widetilde{\mathrm{HEM}}_{i,t} = \mathrm{HEM}_{i,t} - \overline{\mathrm{HEM}}_i,$$

$$\widetilde{R}_{i,t} = R_{i,t} - \overline{R}_i.$$

These measure how far an event deviates from that stock's typical historical expected move and realized earnings move, respectively.

To begin this process its worth exploring the relationship between HEM and the realized move across all eligible earnings events in the data. This would establish the direct correlation between past historical earnings moves and future performance. Correlating each event, where there is an eligible trailing 8-quarter HEM, against its realized move produces a Spearman result of $\rho = 0.318$. This is a modestly positive correlation, but unfortunately this result is contaminated by the overlap of the following concepts: (1) between-stock variation and (2) within-stock variation. The first variation refers to inherent differences in volatility post-earnings between stocks, while the second variation refers to whether a specific stock's current earnings will be larger or smaller than its own normal move. Due to the pooled result having combined both sources of variation one still does not know how much the figure computed is impacted by either factor. In order to isolate the latter factor, which is the variable of interest, one must demean the baseline entirely, leaving only how much each event deviated from that stock's norm. To detect timing a comparison between two event-specific or *rolling* variables must be made. Within the dataset, the only thing which varies event-to-event and is knowable in advance is the rolling:

$$\widetilde{\mathrm{HEM}}_{i,t} = \mathrm{HEM}_{i,t} - \overline{\mathrm{HEM}}_i$$

When the trailing 8-quarter HEM rises above a stock's baseline, it signals that the upcoming earnings announcement appears more volatile than is typical for that stock. To test whether this event-specific signal carries predictive power, each stock's baseline was removed and the correlation recomputed using the residuals. The resulting Spearman correlation of ρ = -0.006 is effectively zero, indicating that once persistent differences between stocks are removed, historical earnings history provides virtually no ability to predict whether a stock's next earnings announcement will deviate from its own typical earnings-move level.

Consequently, the pooled correlation of 0.318 is shown to be driven almost entirely by the durable cross-sectional differences established in Section 5.3, rather than genuine event-level timing information. It's worth noting that the Spearman correlation is an average figure, and therefore not enough to confidently conclude within-stock variation is null across individual stocks. It could very well be the case that history does time events for particular stocks but the positive outcomes are canceled by the group of stocks where history works negatively. An exact conclusion is therefore drawn by computing the correlation inside each stock separately. This yields 469 per-ticker correlations, one for each stock with sufficient earnings history. If history genuinely timed events for a meaningful subset of stocks, these correlations would cluster positively for that group; instead, they center almost exactly on zero, with a mean of 0.037 and only 41.8% positive — indistinguishable from a coin flip. No hidden population of predictable stocks emerges.

A final possibility remains. Although the per-ticker correlations average to zero, they are not identical: some reach +0.4, others 0.4. One might interpret this spread as evidence that history times events for some stocks and not others. However, with only roughly forty earnings events per stock, sampling noise alone produces correlation estimates of this magnitude even when the true relationship is exactly zero. To test whether the observed spread exceeds what noise would generate, the excess-variance ratio is computed — the observed dispersion of the per-ticker correlations divided by the dispersion expected under pure noise. A ratio of one would indicate the spread is entirely attributable to sampling noise; a ratio meaningfully above one would indicate genuine differences in predictability across stocks. The result, 1.05, is effectively one: the apparent variation across stocks is what random noise would produce, and provides no evidence that any stock possesses real event-timing ability.

5.5 Synthesis

All four subsections have arrived at a variety of noteworthy conclusions based on thorough processing and analysis. The former subsections yield a precise characterization of what a stock's own earnings history is able to and unable to tell us. Over 10 years and across 502 securities in the dataset, history describes the *level* of a stock's post-earnings move with confidence; with the typical move being modest (§5.1). On average, systematically history acts as an unbiased forecast of the next realized move (§5.2), and a stock's typical move remains persistent over a decade (§5.3). Unfortunately what history cannot do is predict with virtually any degree of confidence which specific announcement will prove unusually large or small relative to a stock's own norm (§5.4). This asymmetry in outcomes and rather unsettling conclusions are the central results for this section and allow for an ideal transition into the next stage, sharpening the question the remainder of this paper addresses. Given a stock's own history fixes its baseline but provides no useful event-specific timing information, then any forecast that improves on history must rely on something external. A natural place to look is the options market's implied expected move, and Section 6 turns to whether it succeeds.

6 Part 2 - Implied vs. History

History's influence on post-earnings reactions across S&P 500 securities has been well documented in the prior section. The first major question of this report: *how correlated is the options market's implied expected move to a stock's historical realized earnings moves*, is sought to be addressed in this beginning subsection.

6.1 How Heavily do Market Predictions Rely on Stock's Past Behavior?

Over the course of roughly three weeks, from July 21st, 2026 until August 5th, 2026, daily execution of collect_snapshots.py approximately 45 minutes prior to market close have tracked hundreds of securities' ATM straddle prices prior to their earnings release. Across 12 trading days the collection window captures 398 unique tickers, with 156 tickers having T-1/T-0 (1 or 0 days until earnings release) and a clean expiry date (which is defined as less than or equal to three days away). These 156 tickers are of particular interest because the data were collected immediately prior to earnings and near option expiration, when the ATM straddle provides the clearest estimate of the options market's implied expected move. It is across these events that this subsection measures how closely the market's forecast (implied EM) relates to the historical baseline (historical EM).

One thing to note: unlike what was previously defined, for this measurement historical expected move is defined as the mean value of the realized moves across the last 8 quarters. The change in definition arises because historical expected moves are now being compared with the implied expected move, as measured by the ATM straddle price, which is inherently a mean-based estimate. Across all events, historical expected move is plotted on the x-axis and implied expected move on the y-axis. The relationship is then evaluated using (i) Spearman's ρ, (ii) R^2, (iii) the regression slope, and (iv) the intercept. The results when plotting historical EM against implied EM across 133 data points (the pool reduced from 156 to 133 as a handful were dropped due to contaminated expiry dates or a lack of sufficient data to produce an 8-quarter historical EM) are displayed below in Figure 7.

Figure 7 produced some extremely interesting findings. The correlation between the raw data, R^2, is 0.53. While the ranked correlation, Spearman's ρ is 0.62. This suggests a significant, but not overbearing, correlation between implied expected move and historical realized earnings data. This makes sense as the options market tends to rely on a variety of external factors such as news, company guidance, macroeconomic conditions, market sentiment, so on and so forth. The fit slope of 0.79 indicates that for every 1 point up tick in 8-quarter mean HEM, there is a 0.79 increase in implied EM. The fitted linear relationship between historical and options-implied expected moves is given by:

$$\widehat{\mathrm{EM}}_{\mathrm{implied}} = 0.85 + 0.79\,\mathrm{EM}_{\mathrm{historical}}. \quad (5)$$

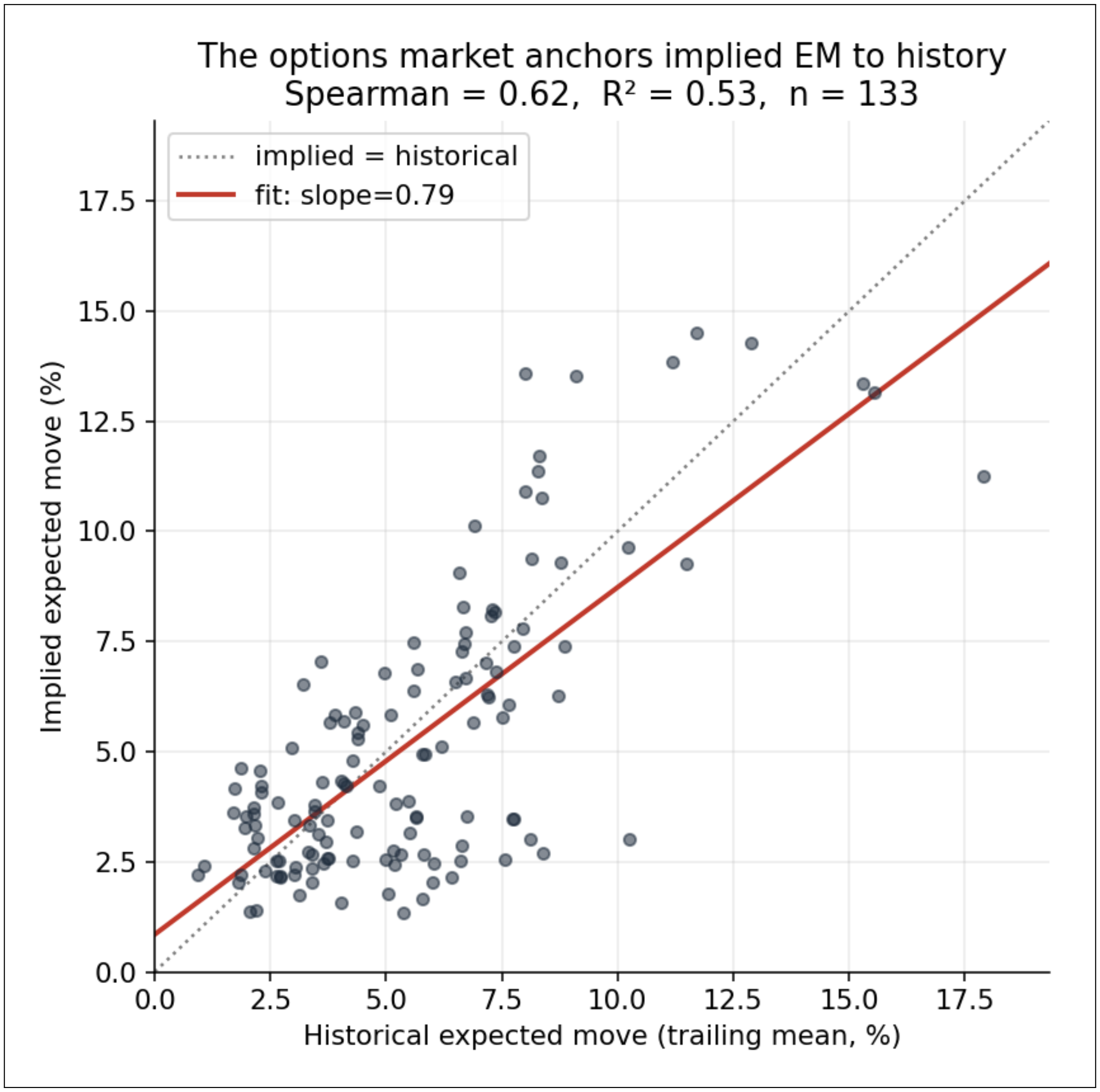

Figure 7: Options market-implied expected move versus the 8-quarter mean historical expected move, with Spearman's ρ, R^2, and the regression slope reported.

The fitted relationship (implied = $0.85 + 0.79 \times$ historical, in percentage points) reveals the following observation: low-historical-EM stocks get priced above their history, while high-historical-EM stocks get priced below their history. The natural question follows: when the market prices a move well above or below what the stock's own history would suggest, what do the post-earnings outcomes look like? The next part of Section 6 addresses this very question and analyzes patterns producing potential market inefficiencies.

6.2 Forecast Accuracy when Historical & Implied Diverge

All 133 events are categorized according to how divergent the implied expected move was relative to the mean 8-quarter historical expected move. There are five quintiles (Q1-Q5) with median divergence ratio figures, ranging from 0.45 to 1.70 (rounded to two decimal places).

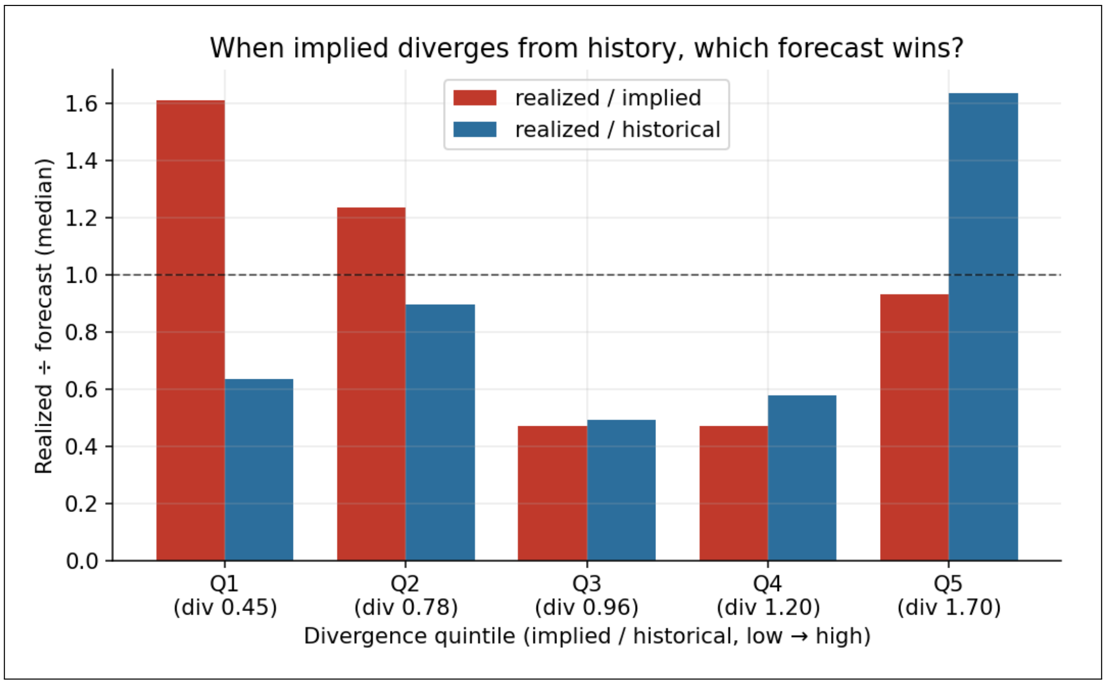

Figure 8: Realized earnings moves relative to implied and historical expected moves across quintiles of implied-to-historical divergence.

The chart reveals a clear pattern of extremes. Among Q5, where the options market priced the expected move the highest (approximately 1.70x) relative to history, the red bar sits at approximately 0.93 (very close to 1). In other words, among the stock pool tested in the last three weeks of earnings releases, the highest divergence ratio resulted in the implied expected move being nearly exact to what actually took place post-earnings. This is direct evidence that implied expected moves incorporate forward-looking, event-specific information that realized history cannot capture – precisely the gap identified in Section 5.4. An interesting observation can be seen in Q1, where the realized outcome lands in between both forecasts: overestimated by historical data and underestimated by market sentiment. The same result occurs in Q2, to a lesser extent. The middle quintiles (Q3, Q4) having divergence ratios of 0.96 and 1.20, respectively, have the realized move coming in below **both** forecasts for both quintiles.

It's difficult to say whether findings presented across 133 distinct earnings events and 12 trading days is any forecast for similar performance in future earnings seasons. With that said, the divergences do carry genuine information. It appears when the market departs upward from history, realized outcomes tend to validate that departure; when it departs downward, the market identifies the correct direction but tends to under-price the magnitude of the residual move.

6.3 Is there a Premium, and How Big?

To close out Section 6, one must consider the practical implications of these findings. Meaning if one were executing a strategy, to say, long or short straddles (or any other appropriate option vehicle of choice), to profit from a post-earnings move based on implied pre-earnings forecasts, the relevant question becomes whether the options market prices that uncertainty accurately relative to what is ultimately realized. In other words, does the options market charge more for an earnings move than what actually materializes? This concept is called *earnings volatility risk premium.*

Across 133 events, the median realized move was 93.5% of the implied expected move – a modest risk premium of approximately 7%. What's worth noting is that if one takes the mean value of the realized/implied ratios across all events, it produces an outcome of 1.135 – meaning realized moves were 13.5% larger, on average. This seems contradictory, but what this indicates is a small number of exceeding realized moves pulls the mean upwards. Furthermore, supporting established market literature, this indicates being short pre-earnings volatility works across majority of earnings events (realized fell below implied 52.6% of the time), but is exposed to potentially severe occasional losses.

7 Limitations

There are quite a few limitations associated with this paper, many of which were stumbled upon as the research process developed. An obvious constraint is the limited stock dataset which the analysis in Section 6 rests upon. The graph correlating implied expected move and historical realized move, as well as the chart displaying the divergence ratios, was based on 133 data points in one earnings season. A more thorough study aimed to increase the accuracy of this research would involve replicating the same process across a greater number of securities and past earnings seasons, thereby strengthening the evidence to support the conclusions drawn, or potentially drawing conclusions different to what this study concluded. Unfortunately this would require access to historical options data — something infeasible during the conduct of this research.

Another restriction is the *survivorship-bias* concept mentioned in Section 4.1. Simply put, this is a logical error which occurs when one only assesses the winners, in a given dataset, and ignores the losers. The S&P 500 equity universe used to source the decade of historical earnings events analyzed in Section 5 excludes companies that were removed from the index during the sample period, whether due to financial distress or other circumstances. Such companies may have experienced large realized reactions, but excluding them takes away from the accuracy of the research, regardless of how disruptive their inclusion may have been to the observed findings.

8 Conclusion

The results of this paper addressed patterns among historical earnings moves and the forecasting potential history has on future earnings events. While also examining the relationship between options pricing based implied expected moves and historical expected moves, a live study was conducted to spot noteworthy patterns and market inefficiencies.

Historical earnings data displays how much a given stock typically moves after earnings. That *typical* move is proven to be consistent over time, solidifying the fact that stocks with volatile post-earnings reactions tend to stay volatile, and those with modest reactions, on average, tend to stay modest. Additionally, systematically over a 10-year period stocks post-earnings moved about as much after earnings as their recent earnings history suggested – within a 4% margin. However, historical data carries virtually no ability to anticipate how a stock will perform post-earnings on an event-to-event basis.

The options market's implied EM is strongly correlated to historical realized moves, but not completely. Implied expected moves are also influenced by forward-looking, event-specific information that history cannot contain. The live study concluded that when implied expectations diverge upward from a stock's historical level, realized outcomes tend to support the direction of that adjustment. At the same time, the market prices a modest 7% earnings-volatility premium for bearing this uncertainty prior to earnings dropping. Simply put, right before earnings announcements, volatility is typically sold at a premium– advantageous to option sellers rather than buyers.

Taken together, the findings portray that the options market efficiently forecasts post-earnings moves by combining historical data with new information, charging a modest premium for the uncertainty. This study is limited to a single earnings season; however the framework developed in this study provides a foundation for extending the analysis into a multi-year study capable of identifying reliable and profitable trading strategies.